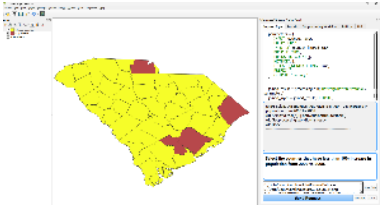
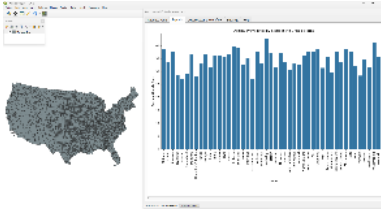
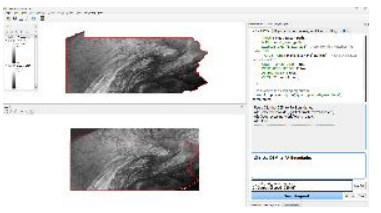
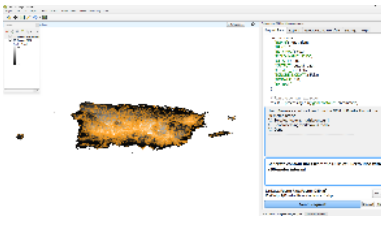
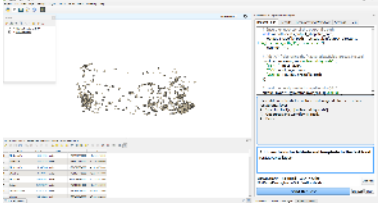
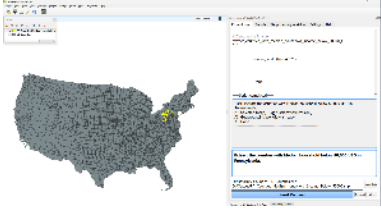
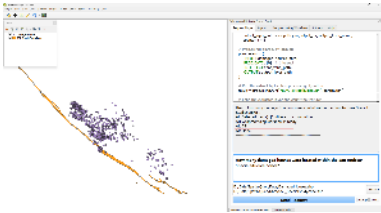
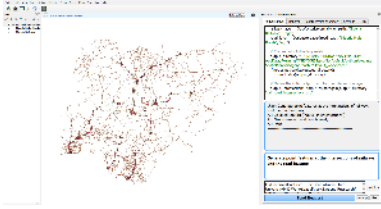
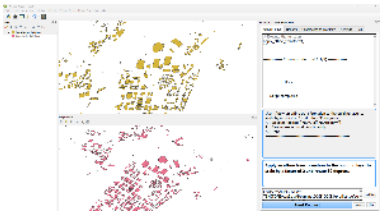
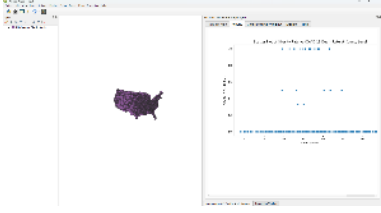


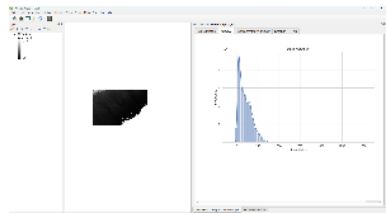
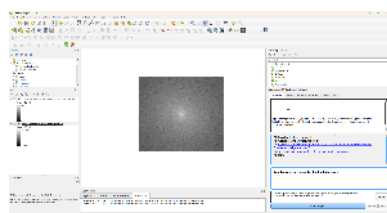
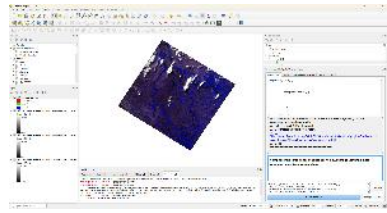
Appendix3: Basic level test cases

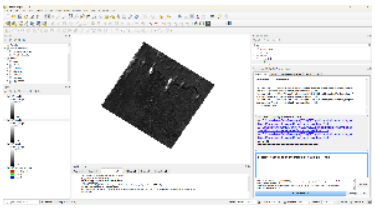
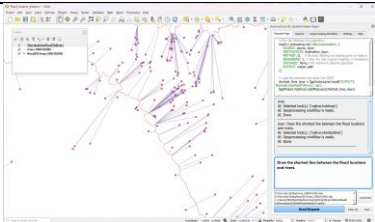
Note that all datasets included here are for testing purposes only.

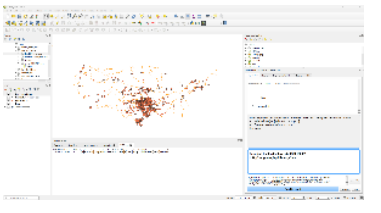
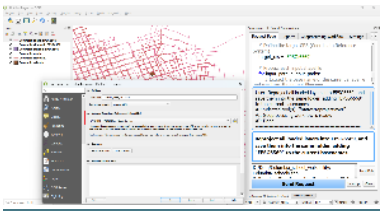
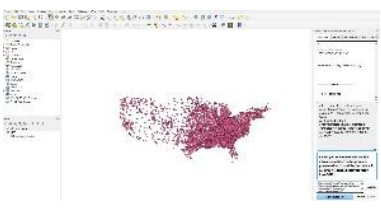
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
1	Select the counties that have less than 10% increase in population from 2005 to 2008.	Vector Selection	native:joinattributable	1	SC_County_CensusData.csv	Population_Increase2005_2008 
2	Create barchart showing obesity prevalence in the United States. Ensure all labels are clear	Plot	seaborn	1	US_County_Obesity	Obesity_Barchart 
3	Clip the DEM to PA Boundaries	Raster Analysis - Clipping	gdal:cliprasterbymasklayer	1	PA_DEM	PA_DEM_Clippping 

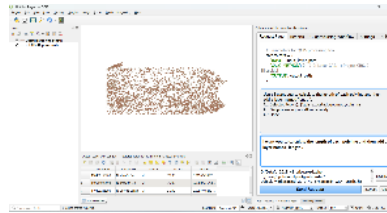
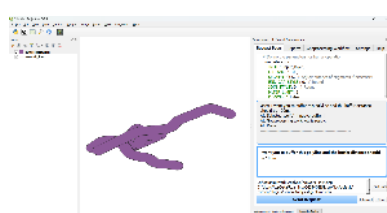
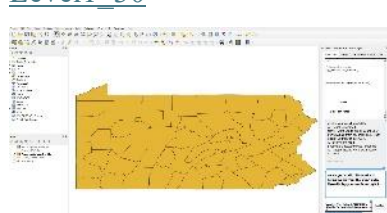
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
4	Generate contour lines from the DEM of Puerto Rico with a 50-meter interval	Raster Analysis - Contour Generation	Gdal:contour	1	Puerto Rico DEM	Puerto Rico Contour 
5	Add new fields for latitude and longitude to the fast food restaurants layer	Add Geometry Attributes	Native:addxyfields	1	PA_Fastfoods	Add XY to Fastfood 
6	Extract the counties with Median household below 50,000 USD in Pennsylvania.	Vector Selection - Attribute Query	qgis:selectbyexpression	1	US_Counties	PA_MedInc below50000 
7	How many damaged houses were located within the San Andreas Special Studies Zones?	Spatial Query	native:extractbylocation	1	DamagedHouses SAF_SpecialStudyZone.shp	DamagedHouses 

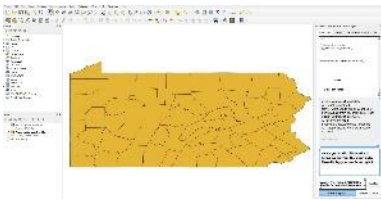
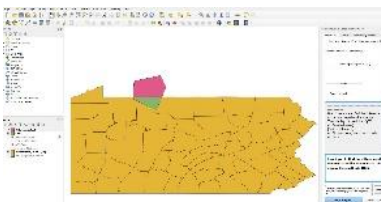
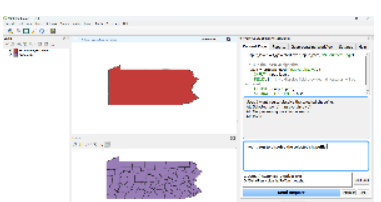
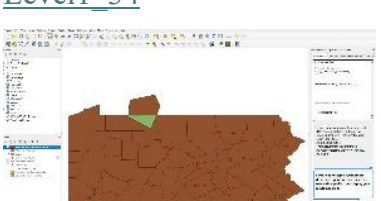
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
8	Generate point features at the intersections of railways and the road features	Vector Overlay Analysis	native:lineintersections	1	Nigeria Major Roads Nigeria Railways	Railway Road Intersection 
9	Apply an affine transformation to the building layer to scale by a factor of 2 and rotate 30 degrees.	Geometric Transformation	native:affinetransform	1	Penn_State_Building	Affine transform 
10	Based on the provided data layers, draw a scatter plot for the Obesity rate and Covid-19 death rate.	Statistical Analysis- plot	seaborn	1	US_Counties_Obesity_Covid	Covid Obesity ScatterPlot 
11	Report the information for this raster data in HTML.	Raster Data Reporting	gdal:gdalinfo	1	SC_DEM	SC_Raster_Data_Report

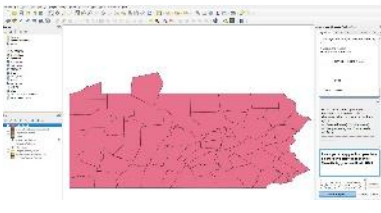
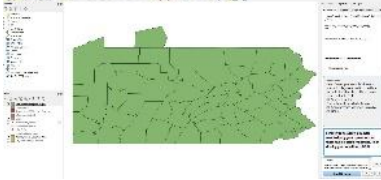
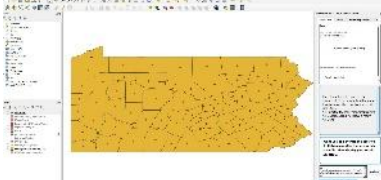
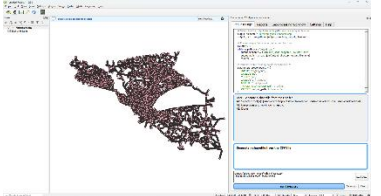
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
						
12	Can you show me the histogram for the DEM pixel values?	Raster Analysis - histogram	seaborn	1	SC_DEM	SC_DEM_Histogram 
13	Do a Fourier conversion for the loaded image.	Image processing	numpy	1	LT05_L1TP_123042_20110928_20200820_02_T1_B5	Fourier1 
14	Composite the three bands of LandSat 5 to a RGB image using the most commonly used band composition.	Image processing	gdal:rearrange_bands	2	TM5_L2/LT05_L2SP_029030_20110605_20200822_02_T1_SR_B2_B3_B4	LandSat_band_composition_success 

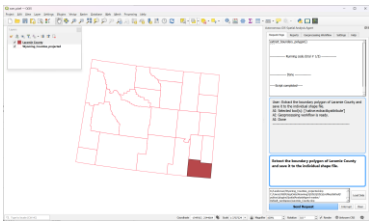
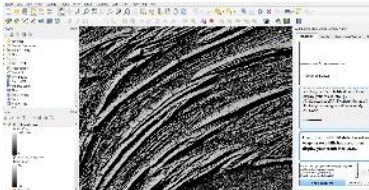
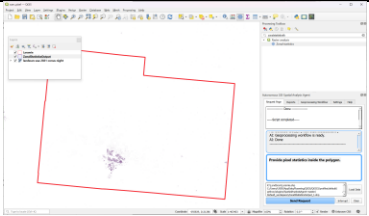
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
15	Separate each band of the image into a single TIF file.	Image processing	gdal:translate	1	composited_rgb.tif	separate_bands 
16	Create a histogram of the distance from the flood event location to its nearest river.	Vector overlay	qgis:distancetonearesthubpoints, seaborn	Failed.	Flood2015may River.shp	
17	Compute the distance of each flood to the nearest river.	Proximity Analysis	['qgis:distancetonearesthubpoints'] ['native:hublines']	Failed	rivers_ESRI102965.shp , flood2015may_ESRI102965.shp	
18	Draw the shortest line between the flood locations and rivers.	Distance Analysis	['native:shortestline']	1	rivers_ESRI102965 , flood2015may_ESRI102965	 flood_nearest_river.png

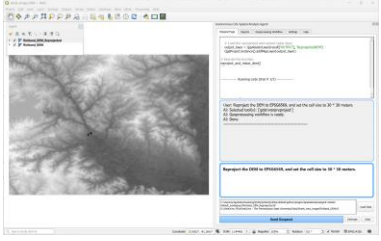
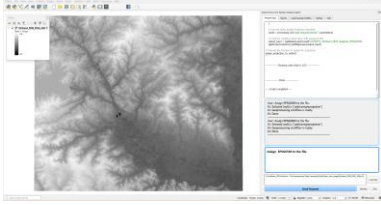
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
19	Reproject the loaded layers to ESRI:102005 USA_Contiguous_Equidistant_Conic.	Reprojection	'gdal:warp' 'reproject'	1	Flood2015may.shp , River.shp	Projection 
20	Reproject all loaded layers into EPSG6569, and save them into the same folder, adding "EPSG6569" to the current basenames.	reprojection	['native:reprojectlayer']	1	Columbia_schools.shp , Columbia_sidewalk.shp , Columbia_roads	Snipaste_2024-11-04_18-09-38.png 
21	I want you to convert this csv file into a shapefile. The longitude is presented as 'X' and the latitude is in column 'Y'.	Data Conversion	['native:createpointslayerfromtable', 'native:savefeatures', 'native:loadlayer']	1	HIV.csv	Level1_25 
22	I want you to only keep US contiguous counties from this input data.	Geographic Filtering	['native:extractbyexpression']	2	US counties shapefile	Level1_26

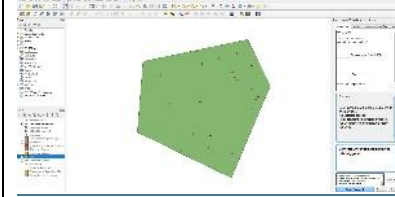
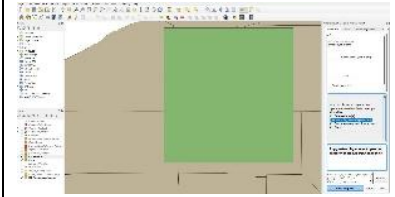
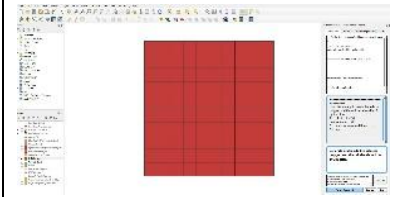
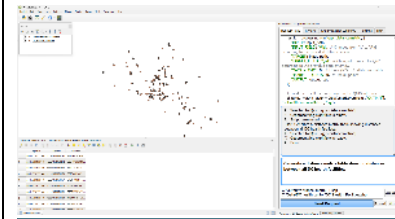
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
						
23	I want you to calculate the length of each polyline and then add a layer named 'length'.	Geometric Calculation	['qgis:exportaddeometrycolumns']	1	tl_2019_42_prisecroads	Length Calculation 
24	I want you to buffer this polyline and the buffer distance should be 0.1m.	Buffering Analysis	['native:reprojectlayer', 'native:buffer']	1	Network_Analysis/network_points	Buffer 
25	I want you to clip all counties of Pennsylvania from this input data.	Clip Analysis	['native:extractbyattribute', 'pdal:clip']	1	US counties	Level1_30 

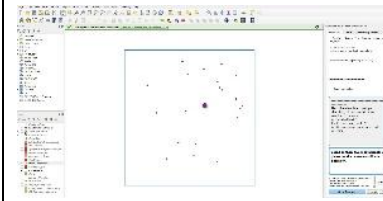
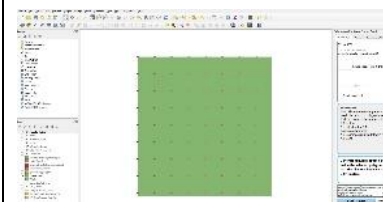
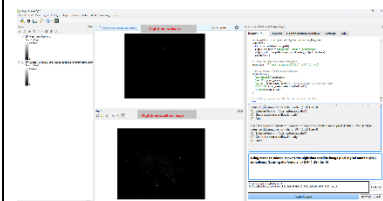
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
26	I want you to identify all points that intersect with Pennsylvania _Counties_Clip data.	Spatial Intersection Analysis	['native:reprojectlayer', 'native:extractbylocation']	1	US counties HIV csv file	Level1_31 
27	I want you to find the difference of the two shapefiles that overlap.	Symmetrical Difference Analysis	['native:difference']	1	generated_polygon , PA counties	Level1_32 
28	I want you to dissolve the selected shapefile.	Dissolve Operation	['native:dissolve']	1	PA counties	Dissolve 
29	I want you to apply symmetrical difference operations on these two selected shapefiles.	Symmetrical Difference Analysis	['native:symmetrical difference']	1	PA counties and generated_polygon	Level1_34 

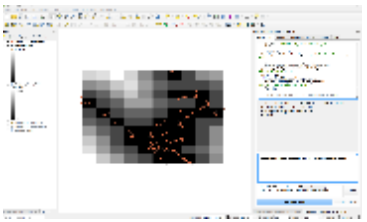
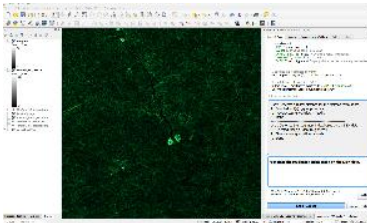
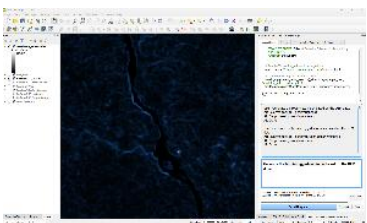
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
30	I want you to apply union operations on these two selected shapefiles. .	Union Analysis	['native:union']	1	PA counties and the generated polygon	Level1_35 
31	I want you to apply eliminate selected polygons operations on these two selected shapefiles.	Polygon Elimination	['qgis:eliminateselect edpolygons']	1	PA counties and the generated polygon	Level1_36 
32	I want you to generate the centriod of all these counties that are in this shapefile.	Centroid Calculation	['native:centroids']	1	PA counties	Level1_37 
33	Generate a shapefile from the CSV file.	Data Conversion	['native:createpointsl ayerfromtable']	2	Street_view_image\ Richland_jsons.csv	CSV to shapefile 

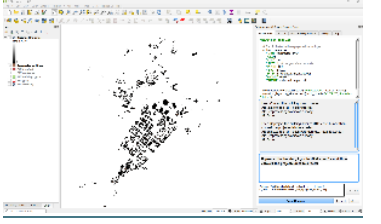
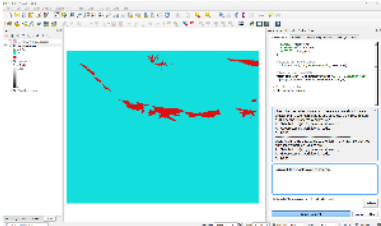
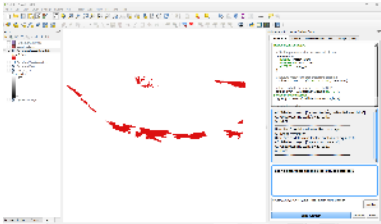
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
34	Extract the boundary polygon of Laramie County and save it to the individual shape file.	Boundary Extraction	['native:extractbyattribute']	1	Wyoming_Counties_projected	Extract_save_feature.png 
35	From this input DEM data, I want you to generate a hillshade	Terrain Analysis	['native:hillshade']	1	n40_w078_1arc_v3	Level1_40 
36	Sum all the pixel values inside a Census tract polygon for each tract.	Zonal Statistics	['native:zonalstatisticsfb']	Failed	landscan-usa-2021-conus-night.tif , tract_37_EPSG4326	
37	Provide pixel statistics inside the polygon.	Zonal Statistics	['native:zonalstatisticsfb']	1	LandScan\Laramie.shp , LandScan/landscan-usa-2021-conus-night.tif	Statistic_pixel.png 

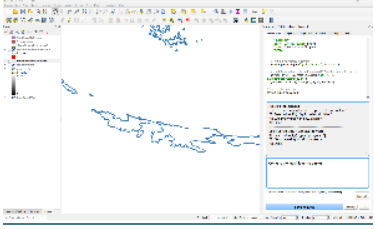
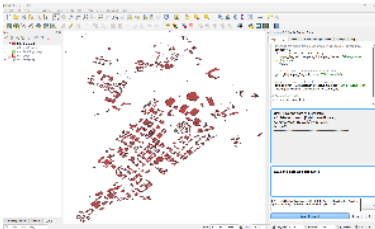
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
38	Reproject the DEM to EPSG6569, and set the cell size to 30 * 30 meters.	Raster Reprojection	['gdal:warp', 'project']	1	Richland_DEM.tif	 DEM_reproject.png
39	Assign EPSG6569 to the file.	Coordinate System Assignment	['gdal:assignproject']	1	Richland_DEM_EPSG6569.tif	 DEM_assign_project.png
40	Find out the number of fastfood restaurants in each county of Pennsylvania	Spatial Join Analysis	native:joinbylocation summary	Failed	PennsylvaniaCounties PA_Fastfoods	
41	Generate 20 random points inside this polygon	Random Point Generation	['qgis:randompointinsidepolygons']	1	Circle.shp	Level1_39

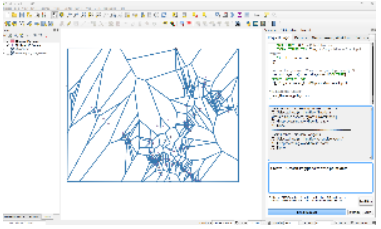
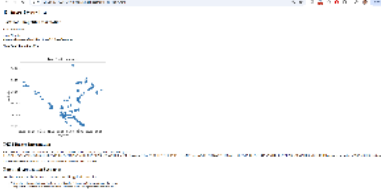
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
						
42	Apply extract layer extent operation based on these two input shapefiles.	Extent Extraction	['native:polygonfrom layerextent']	1	Circle.shp	Level1_40_0 
43	Generate grid inside this selected polygon and the cell size should be 2 by 2 meters.	Grid Generation	['native:creategrid']	1	Rectangle_grid.shp	Level1_41 
44	Generate a distance matrix table showing distances between all DC health facilities.	Distance Matrix Calculation	qgis:distancematrix	1	DC_Health_Facilitie s	Distance_Matrix 

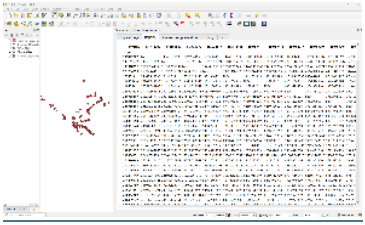
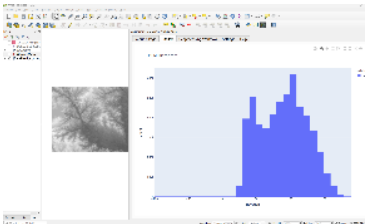
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
45	Based on these two input shapefiles, please conduct mean coordinate operation.	Mean Center Calculation	['native:meancoordinates']	1	Point_inside_polygon.shp ; Rectangle_grid.shp	Level1_43 
46	Identify the 5 nearest neighbors to the center point from the points dataset.	Nearest Neighbor Analysis	['native:joinbynearest']	failed	Points.shp ; Center.shp	
47	Generate regularly spaced points inside the selected polygon, ensuring that the distance between each point is 0.1 meters.	Point Grid Generation	['qgis:regularpoints']	1	Rectangle_grid.shp	Level1_46 
48	Using raster calculator, convert the nighttime satellite image pixel digital number (DN) to radiance (L) using the formula: $L = DN^{1.5} * 10e-10$	Raster calculation.	['qgis:rastercalculator']	1	LuoJial-01_LR201903142986_20190311145630_HDR_0000_gec.tif	Nighttime radiance 

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
49	Generate a raster distance for the fastfood restaurants.	Distance Raster Generation		1	Richland_SC_fastfood.shp	Raster_Distance_Result.png 
50	Generate the roughness index based on the DEM data.	Terrain Roughness Analysis	['gdal:roughness']	1	Richland_SC_DEM.tif	RoughnessIndex_Result.png 
51	Generate the terrain ruggedness index based on the DEM data.	Terrain Analysis.	['gdal:triterrainruggednessindex']	1	Richland_SC_DEM.tif	terrain_ruggedness_index_Result.png 
52	Reproject the building layer to UTM Zone 17 and then	Data Management and Rasterization	['gdal:warp', 'gdal:rasterize']	1	PSU_b.shp	Screenshot 2024-10-24 233849

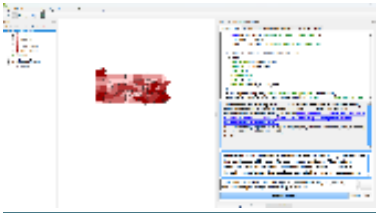
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	convert the projected data to a raster					
53	Use the raster calculator to create a mask showing cells greater than 1000 meters as 1, and cells less than 1000 meters as 0. Use the binary mask expression.	Raster Analysis.	['qgis:rastercalculato r']	1	srtm_41_19.tif	Screenshot.png 
54	Set no data value for the pixel values equal to 0	Raster Data Management.	['qgis:rastercalculato r']	1	elevation_mask.tif	Screenshot 2024-10-24 230407 

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
55	Convert the mask layer to vector.	Data Conversion	['gdal:polygonize']	1	elevation_mask_nodata.tif	Screenshot 2024-10-24 231951 
56	Save the data as a shapefile	Data Export	['gdal:convertformat']	1	elevation_mask_nodata.tif	Screenshot 2024-10-24 233219 

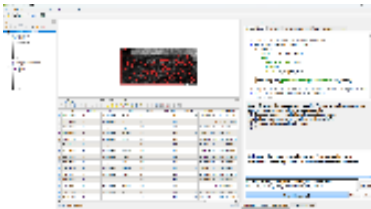
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
57	Create Thiessen Polygons for the point data	Geostatistical Analysis.	['native:voronoipolygons']	1	Richland_SC_fastfood.shp	Screenshot 2024-10-25 001058 
58	Please generate a detailed report in HTML to describe the nature of the two datasets based on your observation. In addition, in the report, please suggest some potential applications or analysis scenarios using the two datasets.	Report Generation	['gdal:gdalinfo', 'native:hillshade']	1	Richland_SC_fastfood.shp Richland_SC_DEM_UTM17.tif	Screenshot 2024-10-25 001856 

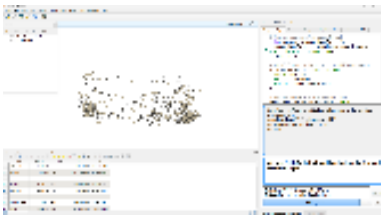
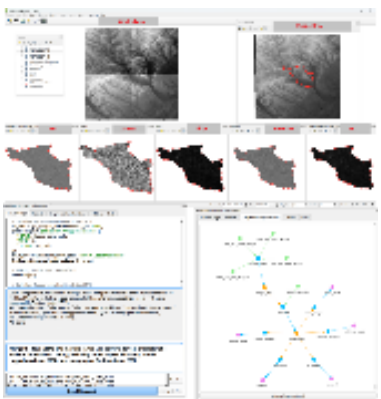
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
59	Generate a HTML table to show the distance matrix in the matrix style.	Matrix Analysis.	['pandas', 'geopandas']	1	distance_matrix.shp	Screenshot 2024-10-25 002825 
60	Please draw a histogram for the DEM data using plotly	Statistical Analysis	['Rasterio', 'NumPy', 'Plotly']	1	Richland_SC_DEM_UTM17.tif	Screenshot 2024-10-25 003225 

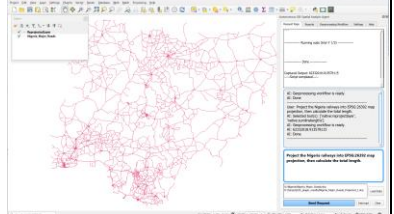
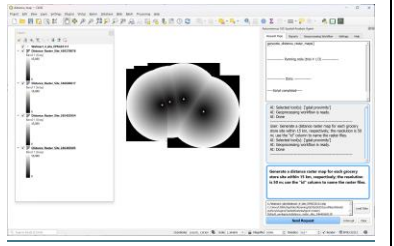
Appendix 4: Intermediate level test cases

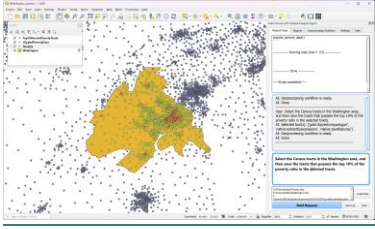
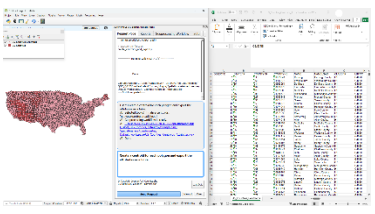
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
1	Perform the following tasks: 1) Count the fast food restaurants in each county and store the result in a new field named "Count". 2) Calculate the fast food accessibility score for each county as $(\text{Count} / \text{Population}) * 1,000$ and store the result in a new field named "Score". 3) Create a thematic map showing the fast food accessibility score for each county.	Spatial Analysis	native:countpointsin polygon, thematic_map_creation	1	/Pennsylvania/PA_Fastfoods_XY.shp /Pennsylvania/PennsylvaniaCounties.shp	Food Accessibility 
2	Perform the following task: 1) Clip the DEM to SC	Terrain Analysis	gdal:cliprasterbymasklayer	1	SouthCarolinaData	SC_DEM Figure

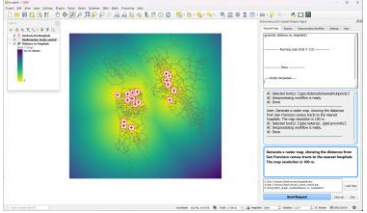
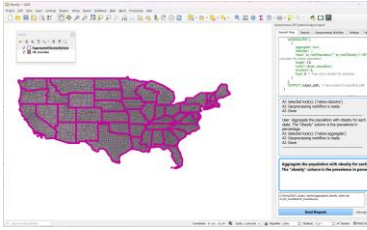
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	boundary and load the clipped DEM. 2) Show me the histogram for the clipped DEM pixel values. 3) Generate the zonal statistics for every county in SC, focusing on average elevation. 4) Finally create a choropleth map showing the average elevation of each county in SC.		seaborn native:zonalstatistics fb			
3	Can you please generate a HTML report to show the building area for each building? You can look at the columns information, if no building area exists, please calculate yourself. Also make sure to use map	Attribute Management	native:reprojectlayer qgis:exportaddgeometrycolumns	1	Penn_State_Buildings.shp	Building report 

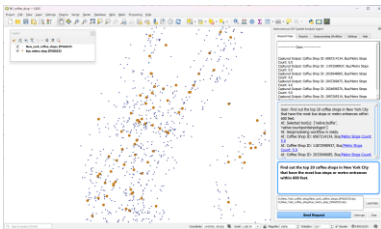
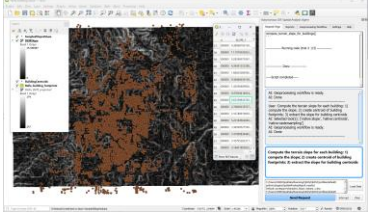
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	projection when calculating area					
4	Perform the following task: 1) Reproject this road vector layer to NAD27/ UTM Zone 10 . 2) Create a new field that contains the length of the road features. Let the name of the field be "Length". 3) Use the 'Length' column and find how many roads have length shorter 100-meters .	Geometric Analysis	gdal:warp native:fieldcalculator native:extractbyattribute	1	SanFrancisco_Roads	Road_Length 
5	Calculate the ruggedness index for Pennsylvania, then summarize the ruggedness index values for each counties	Terrain Analysis	['gdal:triterrainruggednessindex', 'gdal:cliprasterbymasklayer', 'native:zonalstatisticsfb']	1	Pennsylvania/PA_DEM Pennsylvania/PennsylvaniaCounties	PA_Ruggedness_index 

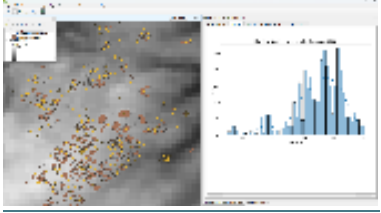
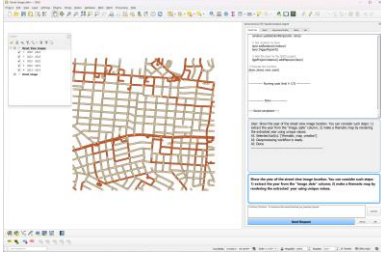
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
6	Add new fields for latitude and longitude to the fast food restaurants layer, then extract the elevation values from the DEM layer and save the results in a new layer.	Data Management	['native:addxyfields', 'native:rastersampling']	1	Pennsylvania/PA_Fastfoods Pennsylvania/PA_DEM	Add XY_to_Fastfood 
7	Merge the four DEMs into a single raster and perform terrain characteristic analysis for Richland County, including slope, aspect, hillshade, terrain ruggedness index (TRI), and topographic Position Index (TPI).	Raster Analysis	pdal:merge, gdal:cliprasterbymasklayer, native:slope, native:aspect, native:hillshade, gdal:triterrainruggednessindex, gdal:tpitopographicpositionindex	1	Richland	Richland_Raster_Analysis.png 
8	Count the number of hospitals and number of fastfoods in each county and	Attribute Query and Data Management	native:joinbylocation summary	2	Pennsylvania/PA_Hospital	PA_FF_HospitalCount

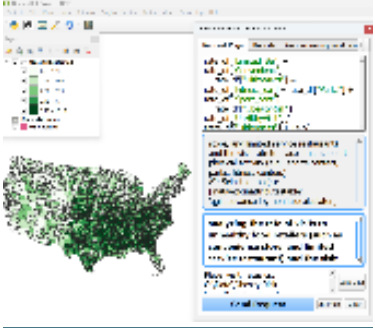
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	save the results into a new county layer by adding two fields.				PA_Counties PA_Fastfoods	
9	Reproject this hospital layer to UTM Zone 17, and then create a 1000 meters buffer around each hospital	Geometric Transformation and Buffering Analysis.	'gdal:warp', 'native:buffer',	1	Pennsylvania/PA_Hospital	PA_Hospital_Buffer 
10	Project the Nigeria railways into EPSG:26392 map projection, then calculate the total length.	Geometric Transformation and Measurement Analysis.	['native:reprojectlayer', 'native:sumlengths']	2	Nigeria/Nigeria_Major_Roads.shp	 Railway_length
11	Generate a distance raster map for each grocery store site within 15 km, respectively; the resolution is 50 m; use the "id" column to name the raster files.	Distance Analysis and Raster Generation		3	Walmart_site/Walmart_4_sites_EPSG32111.shp	Distance_map.png 

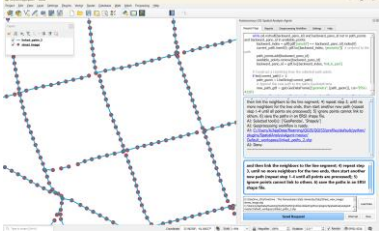
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
12	Select the Census tracts in the Washington area, and then save the tracts that possess the top 10% of the poverty ratio in the selected tracts.	Attribute Query and Statistical Analysis	['native:extractbyexpression', 'pdal:clip']	2	PovertyData	Washington_poverty.png 
13	Create a centroid for each polygon and export the attributes as a table.	Geometric Operations and Data Export	['native:centroids', 'native:exporttospreadsheet']	1	US_counties	Table_Result 
14	Conduct a hospital accessibility analysis following these steps: 1) find out the nearest hospital for each Census tract. 2) compute the distance between each tract's centroid and its nearest	Network Analysis and Distance Analysis	'native:nearestneighborhouranalysis', 'native:centroids', 'qgis:distanceonnearesthubpoints', 'gdal:rasterize', 'gdal:proximity'	Failed	SanFrancisco_tracts_central, SanFranciscoHospitals	

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	hospital. 3) create a raster, indicating the distance from each tract to its nearest hospital.					
15	Generate a raster map, showing the distances from San Francisco census tracts to the nearest hospitals. The map resolution is 100 m.	Distance Analysis and Raster Generation.	['gdal:rasterize', 'gdal:proximity']	3	San_Francisco/SanFranciscoHospitals.shp , San_Francisco/SanFrancisco_tracts_central.shp	 hospital_distance.png
16	Aggregate the population with obesity for each state. The "obesity" column is the prevalence in percentage.	Statistical Analysis and Data Aggregation.	['native:aggregate']	3	US_Counties/US_Counties.shp	 Obesity_population.png

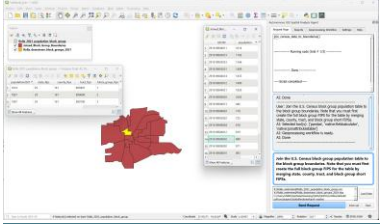
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
17	Find out the top 20 coffee shops in New York City that have the most bus stops or metro entrances within 600 feet.	Proximity Analysis and Spatial Query	['native:buffer', 'native:countpointsin polygon']	3	New_York_coffee_shops\bus_metro_stop_EPSG6535.shp , New_York_coffee_shops\New_york_coffee_shops_EPSG6535.shp	NY_coffee.png 
18	Compute the terrain slope for each building: 1) compute the slope; 2) create centroid of building footprints; 3) extract the slope for building centroids.	Terrain Analysis and Geometric Operations	['native:slope', 'native:centroids', 'native:rastersampling']	3	Rolla_building_footprints , Rolla_DEM_project.ed	Rolla_building_slope.png 
19	Generate centroids for each building feature, extract the elevation values from the	Geometric Operations and Data Extraction	Native:centroids, native:resampling, seaborn	1	Pennsylvania/PSU_DEM	Penn State Building Elevation

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	DEM at each building centroid, and export the elevation data to draw a histogram with 50 bins using Seaborn				Pennsylvania/Penn State Buildings	
20	Show the year of the street view image location. You can consider such steps: 1) extract the year from the "image_date" column; 2) make a thematic map by rendering the extracted year using unique values.	Data Extraction and Thematic Mapping	['native:calculateexpression', 'thematic_map_creation']	3	Street_view_image\street_image.shp	 Street_image_date.jpg
21	Generate an obesity risk behavior index of each county in the USA by analyzing the rate of visits to unhealthy food retailers (such as convenience store, alcoholic drinking places,	Index Creation and Thematic Mapping	native:joinattributable, qgis:advancedpython fieldcalculator, thematic_map_creation	2	US_Counties Obesity_Risk_Score/Place_visit_rates	Obesity_Risk_Figure_New New

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	and limited service restaurant) and the visit rate to places that support physical activity (e.g., sports centers, parks, fitness centers). Visualize the results in a thematic map to highlight the obesity risk behavior index across counties.					
22	You need to compute the location error between the estimated tree trunk and the ground truth: 1) for each ground truth location, find out its nearest estimated location within 3 meters; the nearest distance is the error. 2) compute the average error for ground truth trunks	Accuracy Assessment and Distance Analysis	['qgis:distancetonearesthubpoints']	Failed.	Street_view_image/trunk_location_estimated_EPSG6569.shp, Street_view_image/trunk_location_ground_truth_EPSG6569.shp	

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	that have the associated estimated location.					
23	Link the street view image location points into a path. The image was obtained using a mapping vehicle. All points represent the location of a street view image. Each image has an identifier in "panoId" column; the column "link_f_pan" indicates the panoId of the neighbor image in the forward direction, the column "link_b_pan" indicates the panoId of the neighbor image in the backward direction. You can link the location points by: 1) randomly pick up a point;	Network Analysis and Data Integration	['GeoPandas', 'Shapely']	3	Street_view_image\street_image.shp	 path linking

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	2) extract its forward and backward neighbors, link them together to form a line segment; 3) find the neighbors of the two ends of the segment, and then link the neighbors to the line segment; 4) repeat step 3, until no more neighbors for the two ends, then start another new path (repeat step 1-4 until all points are processed); 5) ignore points cannot link to others. 6) save the paths in an ERSI shape file.					
24	Join the U.S. Census block group population table to the block group boundaries.	Data Join and Attribute Management	['pandas', 'native:fieldcalculator',	1	Rolla_watershed/Rolla_2021_population_block_group.csv ,	Attribute_join.png

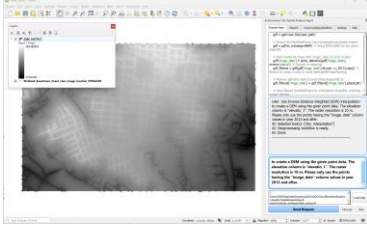
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	Note that you must first create the full block group FIPS for the table by merging state, county, tract, and block group short FIPSs.		'native:joinattributable']		Rolla_watershed/Rolla_downtown_block_groups_2021.shp	
25	Allocate the Census block group population to the building footprint according to the building area. For example, in a block group of 30 residents, there are two buildings with areas of 100 square meters and 200 square meters, respectively; you need to allot 10 residents and 20 residents to the two buildings accordingly.	Data Interpolation	['native:joinattributes', 'qgis:exportaddgeometrycolumns', 'native:fieldcalculator']	Failed.	Rolla_watershed/Rolla_building_footprints.shp , Rolla_watershed/Rolla_downtown_block_groups_2019.shp	

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
26	Use the county boundary to clip the land cover raster for each county in Wyoming State, US. These are steps for your reference: 1) Select the county boundary of a county. 2) Clip the land cover raster using the county boundary. 3) Save each county raster as TIF. Use "NAME" column as the file name.	Raster Clipping and Data Export	['native:extractbyattribute', 'gdal:cliprasterbymasklayer', 'gdal:translate']	3	Landcover\US_Counties_projected.shp, Landcover\nlcd_2021_land_cover_148_20230630.img	 county_clip_landcover.png
27	Clip the image using the extent box of the shape file using these steps: 1) generate the extent box of the shape file; 2) clip and save the image.	Raster Clipping and Geometric Operations	['native:extenttolayer', 'gdal:cliprasterbyextent']	Failed.	NoiseMap\Columbia_image_points_ESRI102039.shp,	

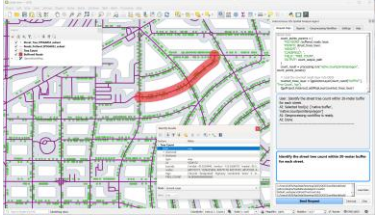
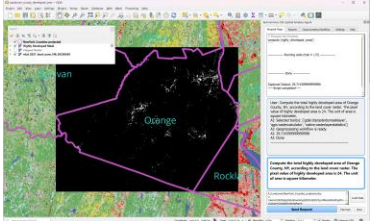
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
					NoiseMap/SC_road_noise_2018.tif	
28	North Carolina will be receiving funding from a car company for building electric charging stations. A group of experts have been asked to determine the factors and weights for locating the new EV charging stations in NC. These experts have determined the most important factors in ranking candidate locations for these new facilities (Data.csv). These experts have also assigned weights to each of the factors. Complete the	Weighted Analysis and Decision Support	['pandas']	1	Data.csv	An HTML output

S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	nominal group process technique by calculating the final weights in the csv file on a 0 to 1.0 point scale using the experts factors and their individual weights.					
29	Using the input DEM layer, generate a watershed with the outlet point specified by the 'Outlet' shapefile. Follow these steps to complete the analysis: 1. Apply a fill operation on the DEM to remove sinks and ensure proper flow.	Hydrological Analysis and Watershed Modeling	['native:fillnodata', 'native:aspect', 'grass7:r.watershed', 'qgis:rastertcalculator', 'grass7:r.stream.extract']	Failed	n40_w078_larc_v3.tif Outlet.shp	

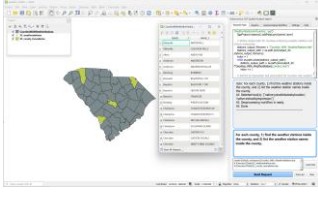
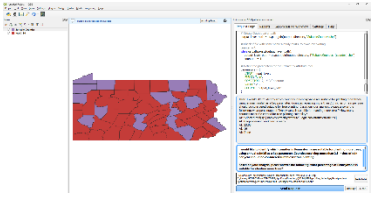
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	2. Perform a flow direction analysis based on the filled DEM data. 3. Calculate the watershed using the flow direction output and the specified outlet point. 4. Run a flow accumulation analysis based on the flow direction output. 5. Finally, generate streamlines from the flow accumulation output, considering only areas with flow					

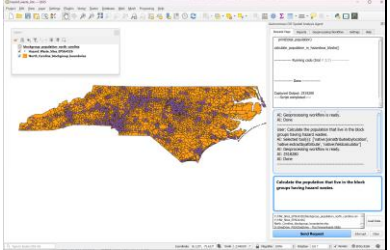
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
	accumulation values greater than 200.					
30	Use Inverse Distance Weighted (IDW) interpolation to create a DEM using the given point data. The elevation column is "elevatio_1". The raster resolution is 10 m. Please only use the points having the "image_date" column values in year 2013 and after.	Spatial Interpolation and Raster Generation	['idw_interpolation']	1	Street tree/Richland downtown street view image location EPSG6569.shp	 IDW.png

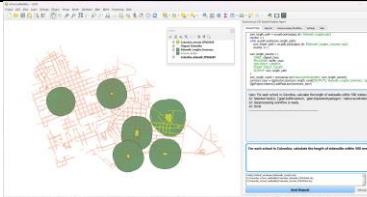
Appendix 5: Advanced level test cases

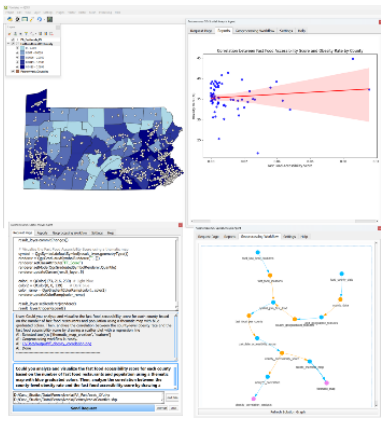
S/N	Task	Category	Tool(s) used	Number of attempts	Dataset used	Screenshot
1	Identify the street tree count within 20-meter buffer for each street.	Proximity Analysis and Spatial Query	['gdal:buffer', 'native:joinbylocationsummary']	2	Street_tree/Street_Tree_EPSG6852 Street_tree/Roads_Potland_EPSG6852_subset	 Tree_count.png
2	Compute the total highly developed area of Orange County, NY, according to the land cover raster. The pixel value of highly developed area is 24. The unit of area is square kilometer.	Area Calculation and Raster Analysis	['gdal:cliprasterbymasklayer', 'qgis:rastercalculator', 'native:rasterlayerstatistics']	3	nlcd_2021_land_cover_I48_20230630.tif NewYork_CountiesProjected	 Developed_area.png
3	Compute the road lengths inside each protected region. The identifier of projected region is "fid".	Spatial Analysis and Length Calculation	['native:intersection', 'qgis:exportaddgeometrycolumns', 'native:aggregate', 'pandas', 'tabulate']	Failed.	Multicriteria Overlay Analysis/roads_polyline.shp	

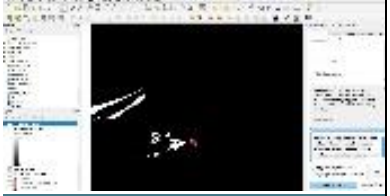
			['native:joinbylocationsummary', 'native:exporttospreadsheet']		Multicriteria Overlay Analysis/Protected region.shp	
4	From PA_Counties dataset select only those counties that the number of data from PA_DATA_EPSG4326 are completely within their boundaries but the count of point is less than 10. Then extract these counties and load them into QGIS.	Spatial Query and Data Extraction	['native:countpointsinpolygon', 'native:extractbyattribute']	1	PA counties shapefile ; PA_Data_EPSG4326	Level3_4 
5	I want you to measure the length of boundary of each county in km based on this input data. Then display your result in an HTML file.	Boundary Measurement and Data Export	['qgis:exportaddgeometrycolumns', 'Pandas library']	1	PA counties	PA_Counties_BoundaryLengths.png 
6	I want you to find all roads that crossed the center county and	Spatial Query and Data Extraction	['qgis:selectbyattribute',	2	TL 2019 42 PRISE CROADS	Level3_6

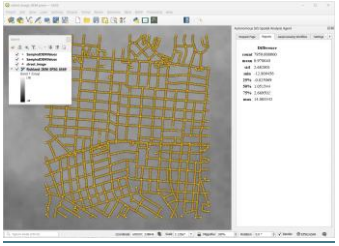
	then extract them from tl_2019_42_prisecroads.		'native:extractbyattribu te', 'native:extractbylocati on']			
7	For each county, 1) find the weather stations inside the county, and 2) list the weather station names inside the county.	Spatial Query and Data Aggregation	['native:joinattributes bylocation', 'native:extractbyattri bute']	2	SC_weatherstations SC_county_boundari es	 weather_station.png
8	I would like to identify which counties in Pennsylvania are suitable for planting more trees, using annual rainfall as a key parameter. Counties receiving more than 2.5 inches of rain per year should be considered suitable for tree planting.	Suitability Analysis and Statistical Assessment	['native:extractbyattr ibute', 'qgis:basicstatisticsfo rfields']	1	Level3_30.shp	Tree_Plalnting_Suitability.p ng 

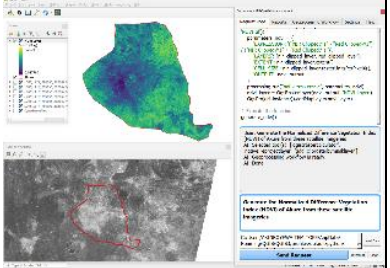
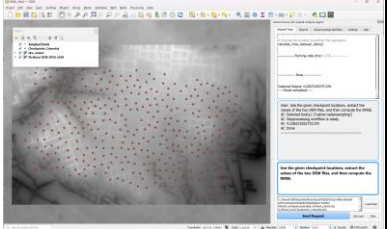
	Based on your analysis, please answer the following: what percentage of Pennsylvania is suitable for planting more trees? How many counties meet the criteria for tree planting suitability?					
9	Calculate the population that live in the block groups having hazard wastes.	Population Analysis and Spatial Query	['native:reprojectlayer', 'native:joinattributesbylocation']	Failed.	HW_Sites_EPSG4326/North_Carolina_blockgroup_boundaries.shp, HW_Sites_EPSG4326/HW_Sites_EPSG4326.shp, HW_Sites_EPSG4326/blockgroup_population_north_carolina.csv	 Hazard_waste.png

10	Based on this input data, identify areas suitable for solar farm installation based on the slope and proximity to power line (RandomPoint). The slope should be less than 5 degrees, and the distance to the nearest power line (RandomPoint) should be within 500 meters.	Suitability Analysis and Proximity Analysis	['native:slope', 'qgis:rastercalculator', 'native:buffer', 'native:polygonize', 'pdal:clip']	2	n40_w078_1arc_v3.tif	Level3_9 
11	For each school in Columbia, calculate the length of sidewalks within 500 meters.	Proximity Analysis	['gdal:buffer', 'gdal:clipvectorbypolygon', 'native:sumlengths']	1	Columbia_school_walkability/Columbia_sidewalk_EPSG6569.shp , Columbia_school_walkability/Columbia_schools_EPSG6569.shp	 school_sidewalk_length

12	I want you to join these two files and create one file that counties contain HIV rates based on this dataset 'HIV Point'. Then generate a symbology map based on HIV rate.	Data Join and Thematic Mapping	['native:joinattributes bylocation', 'thematic_map_creation']	1	HIV shapefile; US counties	Level3_14 
13	Could you analyze and visualize the fast food accessibility score for each county based on the number of fast food restaurants and population using a thematic map with blue graduated colors. Then, analyze the correlation between the county-level obesity rate and the fast food accessibility score by drawing a scatter plot with a regression line	Statistical Analysis and Thematic Mapping	['thematic_map_creation', 'seaborn']	1	PA_Fastfoods_XY, Pennsylvania counties S	Fastfood_accessibility score 

14	Based on this provided DEM data and another observation point (Random_Point), I want you to conduct a viewshade analysis from the observation point. Set a maximum radius for the analysis of 5 kilometers.		Selected tool(s): ['gdal:viewshed']	1	n40_w078_1arc_v3; Random_Point	Level3_26 
15	Please conduct a visibility analysis using the selected datasets. Set the following parameters for the analysis: Observation Offset (Cell Towers): 30 meters. Surface Offset: 1.60 meters. Outer Radius: 2000 meters. After completing the analysis, answer the following questions: How many buildings are visible from a single cellular tower?	Viewshed Analysis and Terrain Analysis	['gdal:viewshed', 'Intersect']	Failed	Viewshade_analysis file	

	What percentage of all buildings are visible from five cellular towers?					
16	Extract the DEM values at the street view images' locations, and then compare them with the elevation in the image's metadata ("elevatio_1" column), print out the descriptive statistics of the difference between the DEM and metadata in an HTML file.	Data Extraction and Statistical Analysis	['native:rastersampling', 'gdal:gdal2xyz', 'grass7:r.univar', 'native:importintopostgis']	1	Street_view_image/street_image.shp , Street_view_image/Richland_DEM_EPSG_6569.tif	street_image_DEM.png 
17	Propose a new Walmart store location, which has similar distances to the existing four locations, while near the roads within 300 meters.	Site Suitability Analysis and Proximity Analysis	['gdal:buffer', 'qgis:distancematrix', 'native:extractwithin', 'native:intersection']	Failed	Walmart_site/Walmart_4_site_EPSG32111.shp , Walmart_site/Motorway_EPSG32111.shp	

18	Generate the Normalized Difference Vegetation Index (NDVI) of Akure from these satellite imageries.	Raster Analysis	['qgis:rastercalculator', 'native:reprojectlayer', 'gdal:cliprasterbymasklayer']	1	Landsat 8 imageries, Akure boundary	NDVI.png  https://drive.google.com/file/d/1NHxrNW1yEDF4naJfbXnh2jJZst0Qv9al/view?usp=drive_link
19	Use the given checkpoint locations, extract the values of the two DEM files, and then compute the RMSE.	Data Extraction	['native:rastersampling']	1	Street_tree/idw_output.tif, Street_tree/Checkpoints_Columbia.shp, Street_view_image/Richland DEM_EPSG_6569.tif	 DEM_RMSE.png
20	Can you help me identify the areas that are suitable for camping. I want the aspect of	Suitability Analysis and	['native:aspect', 'qgis:rastercalculator', 'gdal:rasterize',	Failed	PA_DEM.tif PA_Roads	

	the area to be greater than 100, and the distance to a closest road not exceeding 1000 meters.	Proximity Analysis	'gdal:proximity', 'qgis:rastercalculator']			
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